DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR,
NANOMINIATURE, SINGLE ROW, RECEPTACLE, POLARIZED SHELL,
SOCKET CONTACTS, HORIZONTAL PCB THROUGH-HOLE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32139.

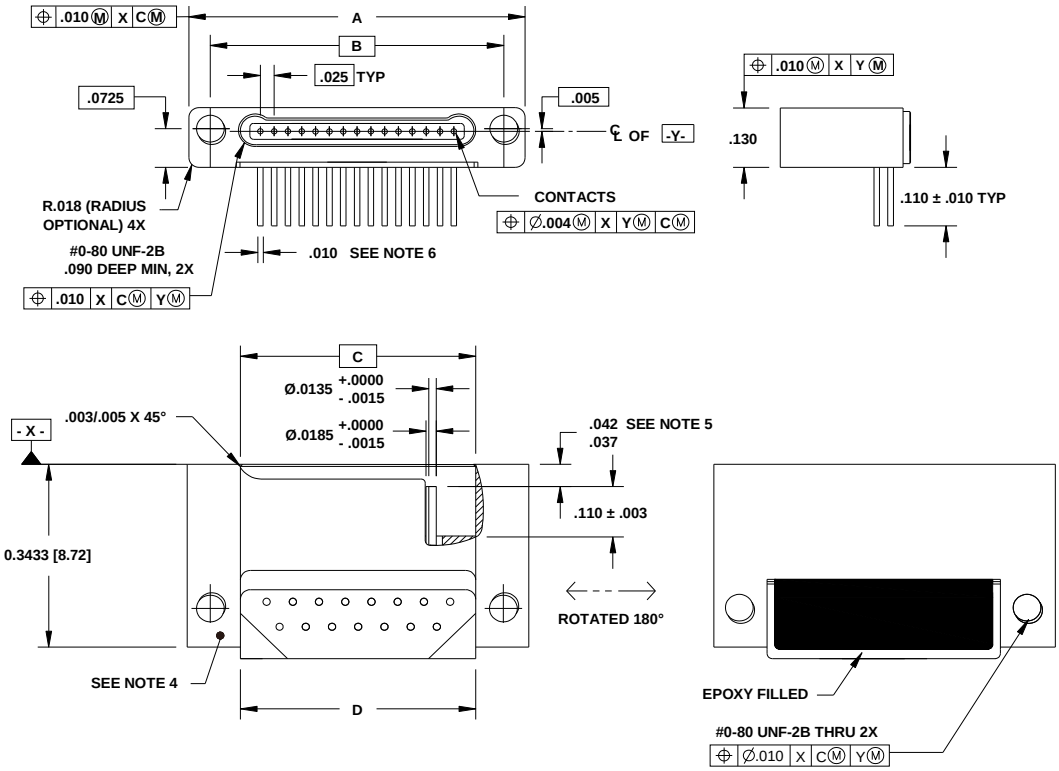


FIGURE 1. Nano connector dimensions and configurations.

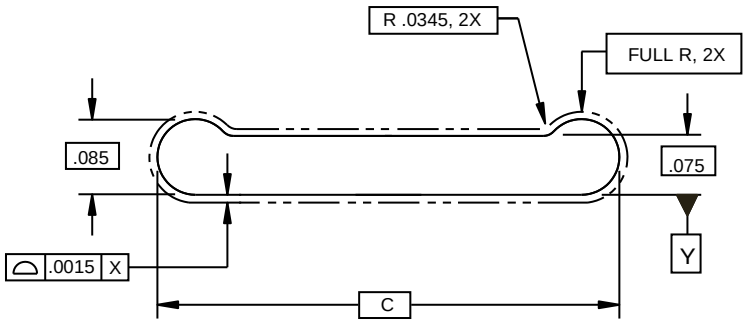
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Insert arrangement	A	B BSC	C BSC	D MAX
9	.500 (12.70)	.395 (10.03)	.285 (7.24)	.295 (7.49)
15	.650 (16.51)	.545 (13.84)	.435 (11.05)	.445 (11.30)
21	.800 (20.32)	.695 (17.65)	.585 (14.86)	.595 (15.11)
25	.900 (22.86)	.795 (20.19)	.685 (17.40)	.695 (17.65)
31	1.050 (26.67)	.945 (24.00)	.835 (21.21)	.845 (21.46)
37	1.200 (30.48)	1.095 (27.81)	.985 (25.02)	.995 (25.27)
51	1.550 (39.37)	1.445 (36.70)	1.335 (33.91)	1.345 (34.16)

Inches	mm	Inches	mm
.0015	0.038	.037	0.94
.003	0.08	.042	1.07
.004	0.10	.090	2.29
.005	0.13	.0725	1.842
.010	0.25	.110	2.80
.0135	0.34	.130	3.30
.018	0.46	.365	9.27
.0185	0.47		
.025	0.63		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm) angular tolerance ±2°.
4. Surface from which the lead length is measured.
5. Shell shall be flush to insulator within ± .004 inch (0.10 mm).
6. Tail finish is tin/lead or gold

FIGURE 1. Nano connector dimensions and configurations – Continued.



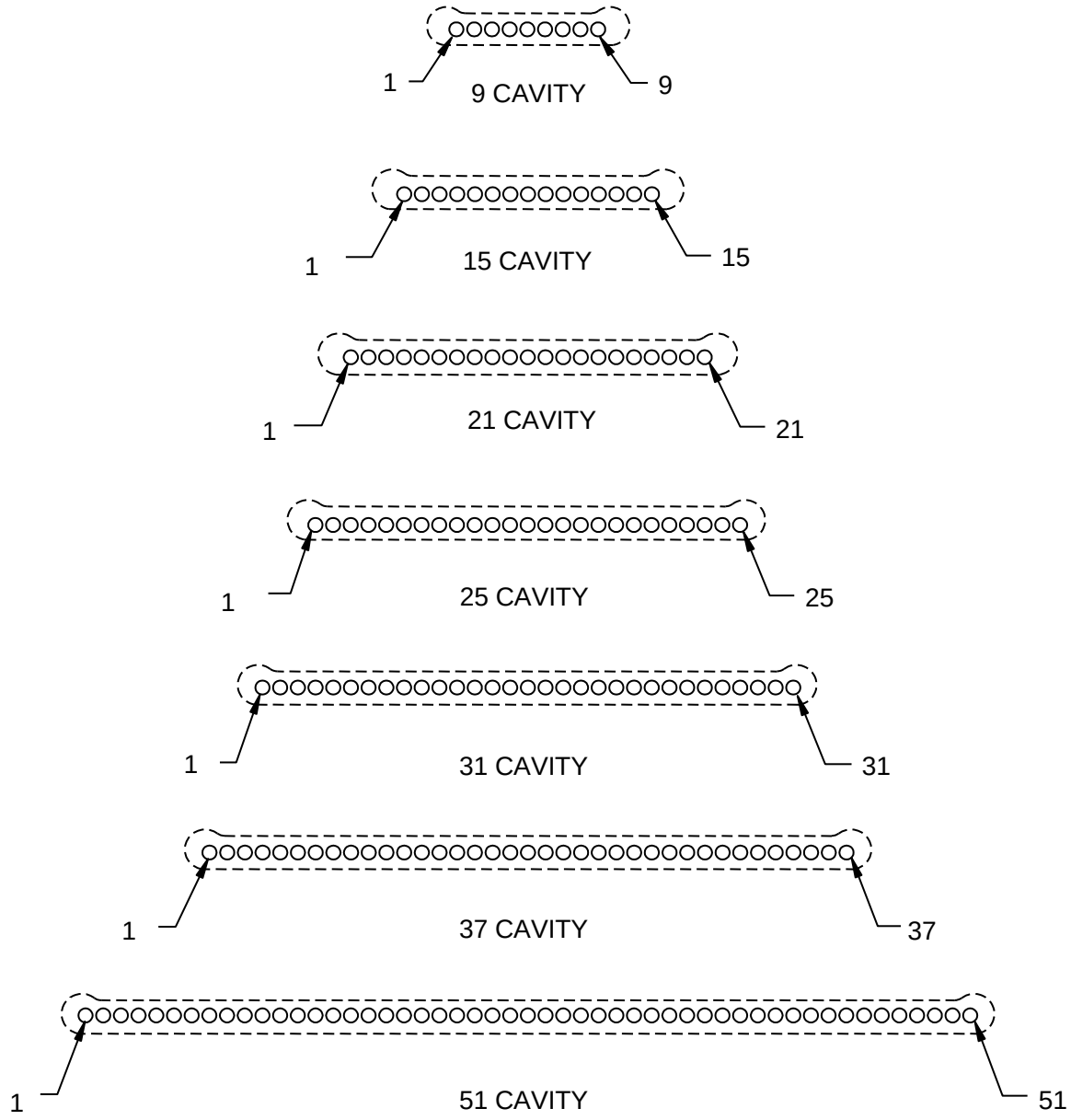
Inches	mm
.0015	0.038
.0345	0.88
.075	1.91
.085	2.16

INTERFACE DETAIL

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm).

FIGURE 2. Nano connector interface.

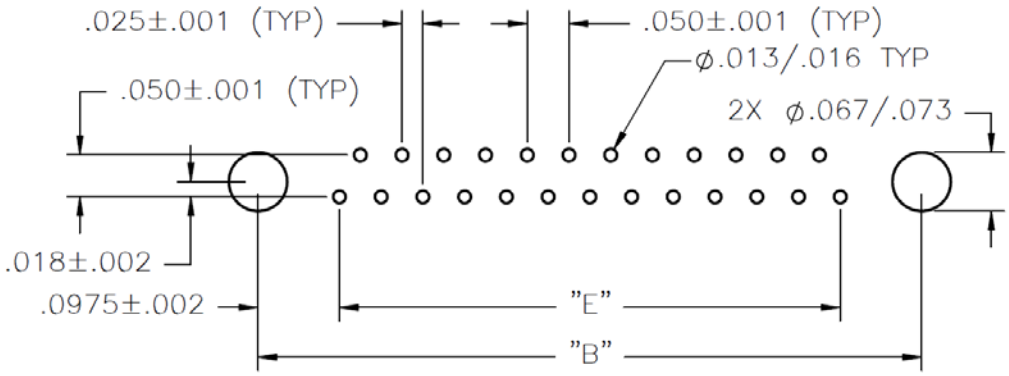
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NOTES:
1. Engaging face of insert shown.
2. Cavity identification numbers are for reference only and do not appear on the part.

FIGURE 3. Nano connector insert arrangement.

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Insert arrangement	B BSC	E
9	.395 (10.03)	.200 (5.08)
15	.545 (13.84)	.350 (8.89)
21	.695 (17.65)	.500 (12.70)
25	.795 (20.19)	.600 (15.24)
31	.945 (24.00)	.750 (19.05)
37	1.095 (27.81)	.900 (22.86)
51	1.445 (36.70)	1.250 (31.75)

Inches	mm
.001	0.02
.002	0.05
.013	0.33
.016	0.41
.018	0.26
.025	0.64
.050	1.27
.067	1.70
.073	1.85
.0975	2.477

NOTES:
1. Reference figure 1 for hole pattern alignment orientation detail.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Unless otherwise specified tolerances are ± .005 inch (.13 mm) angular tolerance ±2°.

FIGURE 4. Nano connector suggested PCB hole pattern.

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REQUIREMENTS

Dimensions and configuration see figures 1, 2, and 3.

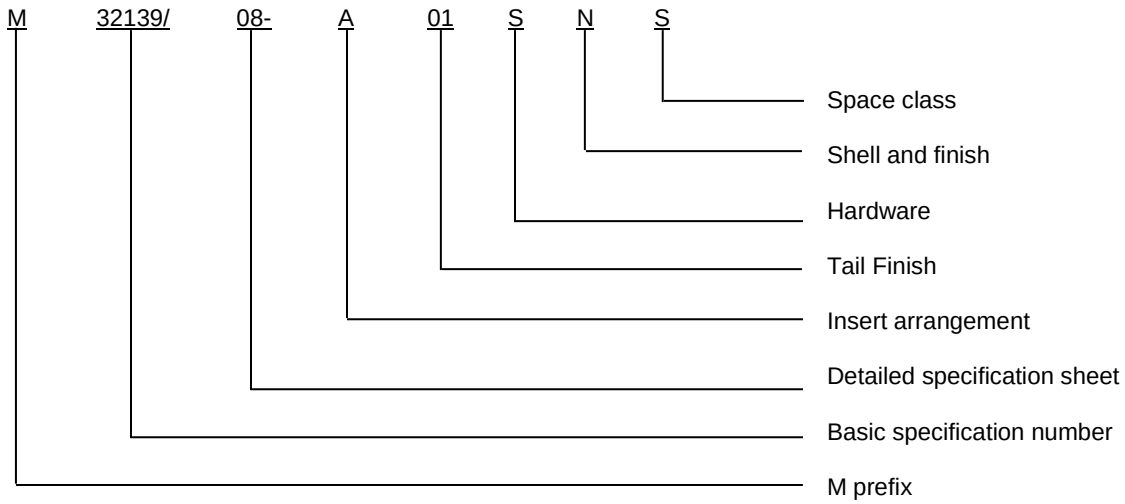
This specification sheet describes the socket side of a rectangular connector. This connector uses reverse gender contacts, i.e., the live pin is recessed in the insulator with the static socket protruding from a shrouded interface.

Contact connection: The socket contact, which is exposed in the insulator, is normally connected to the neutral side of the circuit.

Sockets are terminated to a printed circuit board (PCB) as defined in figure 4.

Mating receptacle: Shall be in accordance with MIL-DTL-32139/1, MIL-DTL-32139/5, and MIL-DTL-32139/7.

Part or Identifying Number (PIN):



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Insert Arrangement	Tail Finish
A = 9	01 = Tin/Lead
B = 15	02 = Gold
C = 21	
D = 25	
E = 31	
F = 37	
G = 51	

Hardware [1/]	Shell and Finish	Space class
T = Thread hole	C = Aluminum shell, cadmium finish	Blank for non-space applications
	N = Aluminum shell, electroless nickel finish [2/]	S = Space class
	S = Passivated stainless steel shell	
	T = Titanium shell	
	A = Pure Electrodeposited Aluminum	
	Z = Zinc Nickel	
	F = Nickel Fluorocarbon Polymer	

1/ Supplied installed.
2/ When aluminum shells are required for space applications, electroless nickel shall be used. Cadmium finish is not acceptable (see MIL-DTL-32139).

Alternate shell finishes:

Pure Electrodeposited Aluminum. Pure dense electrodeposited aluminum shall be in accordance with MIL-DTL-83488, type II shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Nickel Fluorocarbon Polymer. High phosphate nickel with fluorocarbon polymer additive, over suitable underplate shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

Zinc Nickel. Zinc Nickel Alloy shall be in accordance with ASTM B841, over a suitable underplate shall withstand 48 hour salt spray. Color shall be non-reflective and shall meet the requirements as specified herein.

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Referenced documents: In addition to MIL-DTL-32139, this document references the following:

MIL-DTL-32139/1
MIL-DTL-32139/5
MIL-DTL-32139/7
MIL-DTL-83488
ASTM B841

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AV, MI
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-2013-178)

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